introduce clustering, explaining what it is and where it stands with respect to machine learning and AI. We'll see that there are different types of clustering, with radically different approaches, and then we will present and discuss in detail 3 algorithms that uses different approaches; by going through the whole chapter readers will be exposed to the theoretical foundations for this topic, learn about algorithms that can be implemented or just applied to break down datasets into smaller homogeneous groups, and also, in the process, get a deeper understanding of nearest neighbor search and multi-dimensional indexing.

But before we start, let's quickly introduce an example of a problem that motivates the use of clustering. Across the previous chapters included in part 2 of this book we have developed this example about an e-commerce site, starting from the early days when Internet became mainstream; now it's time to bring our company into the years 2010s, and add a data-science team. In fact, to have sales thrive, we will need to perform customer segmentation in order to understand our customers' behavior and categorize customers based on what we know about them[1]: their purchasing habits, their financial situation, but also their demographic, since age, level of education, Country or State where they live are all factors that influence people's taste and spending capacity.

Customer segmentation partitions customer into homogeneous groups sharing similar purchasing power, similar purchase history, or similar expected behavior. Clustering is one step of this process, where the groups are formed from raw, unlabeled data; clustering algorithms don't